Robust Remote Sensing Scene Classification via Diffusion-Based Knowledge Distillation Under Mixed-Noise Degradation

Longwei Li, Xinyu Chu, Xiang Li, and Nan Li

***Abstract*—High-resolution remote sensing (RS) scene classification is pivotal for intelligent Earth observation but faces significant deployment challenges on resource-constrained devices, primarily due to the prohibitive computational cost of state-of-the-art deep models. Knowledge distillation (KD) mitigates this by transferring knowledge from a large teacher model to a compact student model, yet its efficacy is often limited by the representational capacity gap between models and further compromised by the complex, mixed-noise degradation inherent in RS imagery during acquisition and transmission. To address this, we adapt and enhance a diffusion-based KD (DiffKD) framework for the RS domain. Our approach explicitly purifies noisy student features using a diffusion model trained on clean teacher features, thereby facilitating more purified knowledge transfer. We further design a comprehensive hybrid noise model that simulates realistic RS physical degradations (sensor, transmission, compression, and atmospheric noise) for rigorous robustness evaluation. A novel teacher model optimization strategy combining progressive fine-tuning and layer freezing is proposed to bridge the domain gap from natural to RS images and ensure a high-quality knowledge source. Extensive experiments on four public RS scene classification benchmarks (NWPU-RESISC45, AID, EuroSAT, and PatternNet) demonstrate that our adapted DiffKD framework consistently outperforms recent state-of-the-art methods in classification accuracy. Crucially, it exhibits exceptional robustness under varying noise intensities, with the student model even surpassing the teacher's performance. This work provides a practical solution for deploying high-performance, noise-resilient scene classification models on edge devices. Code is available at https://github.com/maomaoxy/DiffKD_2025**

***Index Terms*—Diffusion models, knowledge distillation (KD), model compression, noise modeling, remote sensing (RS), robustness, scene classification.**

I. Introduction

As A fundamental problem in understanding high-resolution remote sensing (RS) imagery, image scene classification plays an important role in the field of Earth observation. Effectively interpreting RS images and extracting informative features remain significant challenges for RS image classification [1], [2]. This technology provides critical information support for a wide array of applications, such as land use and land cover (LULC) mapping, smart city management, natural resource surveys, disaster emergency response, and global change studies [2], [3], [4]. Consequently, it remains a persistent research focus within the RS community. However, significant challenges arise from the unique characteristics of RS data. Compared to conventional natural scene images, RS imagery is more complex. Unlike ground-level image scenes, high spatial resolution RS images may contain various types of ground objects. Owing to the large-scale field of view from a bird's-eye perspective, RS scenes typically exhibit greater complexity and heterogeneity in object distribution and spatial arrangement [5], [6]. This leads to fundamentally different visual patterns and semantic representations. Furthermore, RS scenes exhibit high intraclass variance and low interclass separability. Additionally, RS imagery is susceptible to degradation caused by multiple factors, including seasonal variations, meteorological conditions, resolution discrepancies across different modalities and differences in solar illumination angles [7], [8], [9], [10], [11]. These inherent complexities pose substantial challenges for building robust and accurate scene classification models.

To tackle these complexities, the advent of deep learning has revolutionized the field of RS scene classification. This has precipitated a paradigm shift, moving away from traditional methods reliant on manually designed features (e.g., SIFT, HOG) toward data-driven approaches that automatically extract hierarchical semantic representations [12], [13]. Contemporary deep learning-based methodologies are primarily dominated by two architectural paradigms: Convolutional neural networks (CNNs) and models based on self-attention mechanisms, notably Vision Transformers [14], [15]. CNN-based models, such as VGGNet and ResNet, excel at extracting local spatial features owing to their inherent inductive biases [16], [17] and are consequently widely employed as powerful backbone feature extractors. In contrast, Transformer-based models (e.g., Vision Transformer, Swin Transformer) leverage a global self-attention mechanism. This enables them to effectively capture long-range dependencies within an image,

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demonstrating significant potential for understanding complex scenes [18], [19], [20], [21].

Beyond scene classification, significant progress has also been made in RS image segmentation, which aims at pixel-wise labeling of land cover and objects. Recent advances include U-Net-like transformers (e.g., UNetFormer [22]), multibranch transformer-UNet hybrids (MBT-UNet [23]), and methods that enhance global–local information (GLE-net [24]) or adjust semantic features adaptively (SFA-Net [25]). Fast transformer-based fusion [26] enables efficient segmentation of ultra-high-resolution imagery, while novel architectures based on Mamba [27] and diffusion models (RS-Dseg [28]) have further pushed the state of the art. Moreover, cross-domain adaptation techniques [29] have been developed to improve generalization across different RS datasets. Although these segmentation methods focus on dense prediction, they share underlying deep learning advances with scene classification and offer valuable insights for future extensions of our denoising distillation framework—for instance, to pixel-level tasks.

However, achieving high performance often comes at a cost. Despite their success, both paradigms face significant challenges for practical deployment. Deep CNNs often struggle to effectively model global contextual information. Conversely, Transformer models are frequently burdened by their massive number of parameters and high computational complexity, making efficient inference on resource-constrained edge devices particularly challenging [30], [31]. Beyond architectural limitations, RS imagery is subject to inherent mixed-noise degradation, which arises from a confluence of physical and processing factors. Key sources include sensor noise (e.g., from photoelectric conversion), bit errors during satellite-to-ground transmission, atmospheric scattering and absorption, and artifacts introduced by lossy compression for efficient storage [31], [32], [33]. This complex mixture of noise sources collectively degrades image quality, which in turn directly undermines a model's discriminative power and generalization ability. Traditional two-stage pipelines that perform denoising prior to classification are not only inefficient but also risk error propagation. Furthermore, the efficiency bottleneck in processing massive RS data poses another major constraint. High-resolution RS images cover extensive geographical areas with rich detail, resulting in enormous dataset volumes. Consequently, many computationally intensive deep models struggle to satisfy the stringent latency and resource-efficiency constraints of real-world, practical deployments [34], [35].

In an effort to reconcile model performance with deployment efficiency, knowledge distillation (KD) has emerged as a highly effective technique [36], [37], [38], [39], [40]. The core premise of KD is to transfer the "knowledge" encapsulated within a powerful, pretrained teacher model to a more compact and efficient student model. In the conventional KD framework, this is often achieved by minimizing the Kullback–Leibler (KL) divergence between the softened output distributions (logits) of the teacher and student networks. This approach has proven successful in enhancing the performance of lightweight models [38], [39], [40]. However, a critical limitation arises in the RS context, particularly under mixed-noise degradation. The standard KD paradigm of directly mimicking teacher features or outputs may cause the student to inherit not only beneficial knowledge but also the noise artifacts embedded within the teacher's representations of the corrupted data, a phenomenon known as "noise propagation" in distillation. This phenomenon ultimately limits the student's robustness in noisy environments [40], [41]. The central challenge, therefore, is to facilitate clean and robust knowledge transfer across the capacity gap between teacher and student, especially when the training data itself are degraded.

Addressing this specific limitation requires a mechanism capable of actively purifying features rather than passively mimicking them. Inspired by the remarkable success of diffusion models in generative image denoising tasks [42], [43], we adapt and enhance a diffusion-based KD (DiffKD) framework for RS scene classification. Our key innovation lies in explicitly integrating a diffusion process into the distillation loop. We treat the student's feature representations as a noisy instantiation of the teacher's cleaner features. A diffusion model, trained solely on clean teacher features, is then employed to denoise the student's features during training. This process aims to construct a unified, lightweight framework that jointly performs denoising and discriminative learning. The explicit denoising step is designed to purify the knowledge transfer pathway, thereby significantly enhancing the student model's feature learning capability and robustness under noisy conditions. The framework is shown in Fig. 1.

The principal contributions of this work are threefold: we adapt and rigorously evaluate a DiffKD framework to enhance the robustness of RS scene classification, explicitly addressing domain-specific challenges. First, we introduce a targeted teacher model optimization strategy to bridge the domain gap from natural to RS imagery, ensuring a high-quality knowledge source. Second, we design a realistic hybrid noise model simulating sensor, transmission, compression, and atmospheric degradations, establishing a rigorous and realistic benchmark for evaluating model robustness. Extensive experiments on four benchmark datasets demonstrate that our approach not only surpasses state-of-the-art methods in accuracy but also achieves exceptional noise robustness, enabling a lightweight student model to outperform its teacher under strong noise interference while maintaining efficiency.

The remainder of this article is organized as follows. Section II details the proposed methodology, including the adapted DiffKD framework, the hybrid noise model, and the teacher optimization strategy. Section III describes the experimental setup, covering the datasets, implementation details, and evaluation protocols. Section IV presents and comprehensively analyzes the experimental results. This includes performance comparisons with state-of-the-art methods, robustness analysis under various noise conditions, and ablation studies validating the contribution of each component. Section V provides discussion and potential future work. Finally, Section VI concludes this article by summarizing the main findings.

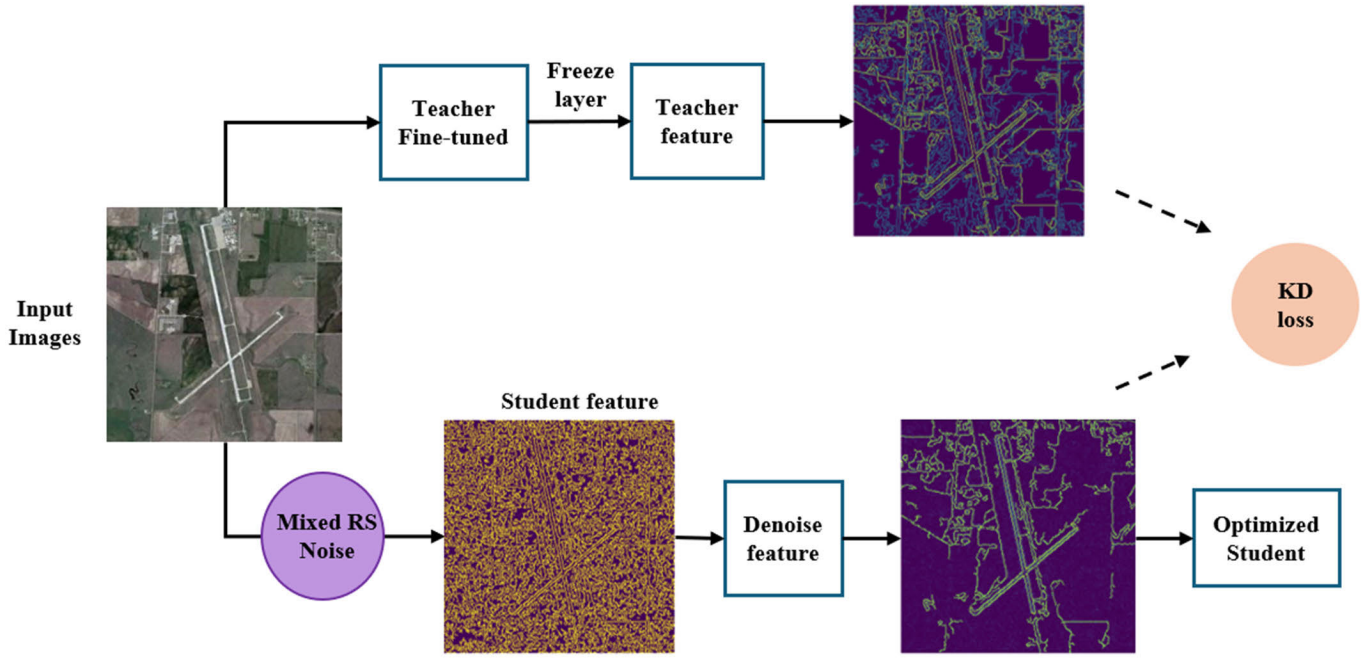


Fig. 1. DiffKD under mixed RS noise degradation.

II. Methodology

This section details our approach to robust RS scene classification. We begin by outlining the adapted DiffKD–framework (Section II-A), which forms the core of our distillation process. Subsequently, we describe the teacher model optimization strategy (Section II-B), the proposed hybrid noise model for RS imagery (Section II-C), and specific adaptations made for handling RS data characteristics (Section II-D).

A. Adapted DiffKD Framework

The efficacy of KD is frequently constrained by the discrepancy between teacher and student feature representations. We formulate a novel perspective: the feature representations extracted by a capacity-limited student model can be conceptualized as a "noisy" or degraded instantiation of the more refined and informative features from the teacher's model. Consequently, direct feature alignment risks propagating these noise artifacts, ultimately limiting the student's robustness. To mitigate this, we integrate a diffusion model as a feature purification module within the KD loop, adapting the DiffKD concept. Fig. 2 illustrates the comparison between the traditional KD framework and our proposed DiffKD-based student model.

The framework operates in two sequential phases: diffusion model training and denoising distillation.

In the first phase, the teacher model's feature maps, $F^{(tea)}$ are used to train a noise prediction network Φ_θ. This is achieved by executing a forward diffusion process that iteratively corrupts the clean features by adding Gaussian noise to $F^{(tea)}$ across time steps t, and training Φ_θ to predict the added noise. The objective function is

$$L_{diff} = \left\| \epsilon - \Phi_\theta \left(F_t^{(tea)}, t \right) \right\|_2^2 \tag{1}$$

where L_{diff} denotes the diffusion training loss. Represents the noise prediction network parameterized by θ. The time step t is uniformly sampled from {1,..., T}, where T is the total number of diffusion steps. $\epsilon \sim N(0, I)$ is the Gaussian noise added to the clean teacher F^{tea} to generate the noisy feature map at step t.

In the second phase, during student training, the extracted student features $F^{(stu)}$ are fed into the *pretrained, fixed* diffusion model Φ_θ. A reverse denoising process is applied over a reduced number of steps (e.g., 5) to obtain a purified student feature estimate $F^{(stu)}$. The final distillation loss is computed between the denoised student features and the original teacher features

$$L_{diffkd} = d\left(\hat{F}^{(\text{stu})}, F^{(tea)}\right) \tag{2}$$

where $d(\cdot, \cdot)$ is a distance metric such as mean-squared error (*mse*). This process encourages the student to learn from a noise-corrected target, leading to more robust feature learning.

B. Teacher Model Optimization and Freezing Strategy

A teacher model pretrained on large-scale natural image datasets (e.g., ImageNet) often suffers from significant performance degradation on RS data due to the substantial domain shift, initially yielding low accuracy (~5%). To ensure a high-quality knowledge source, we implement a two-stage optimization strategy.

Phase 1 (Classifier Warmup): The pretrained backbone weights are kept frozen, while solely the randomly initialized classification head is fine-tuned on the target RS dataset to quickly adapt to the new label space.

Phase 2 (Feature Refinement): After classifier convergence, the final few layers of the backbone are frozen and fine-tuned jointly with the head using a conservative learning rate

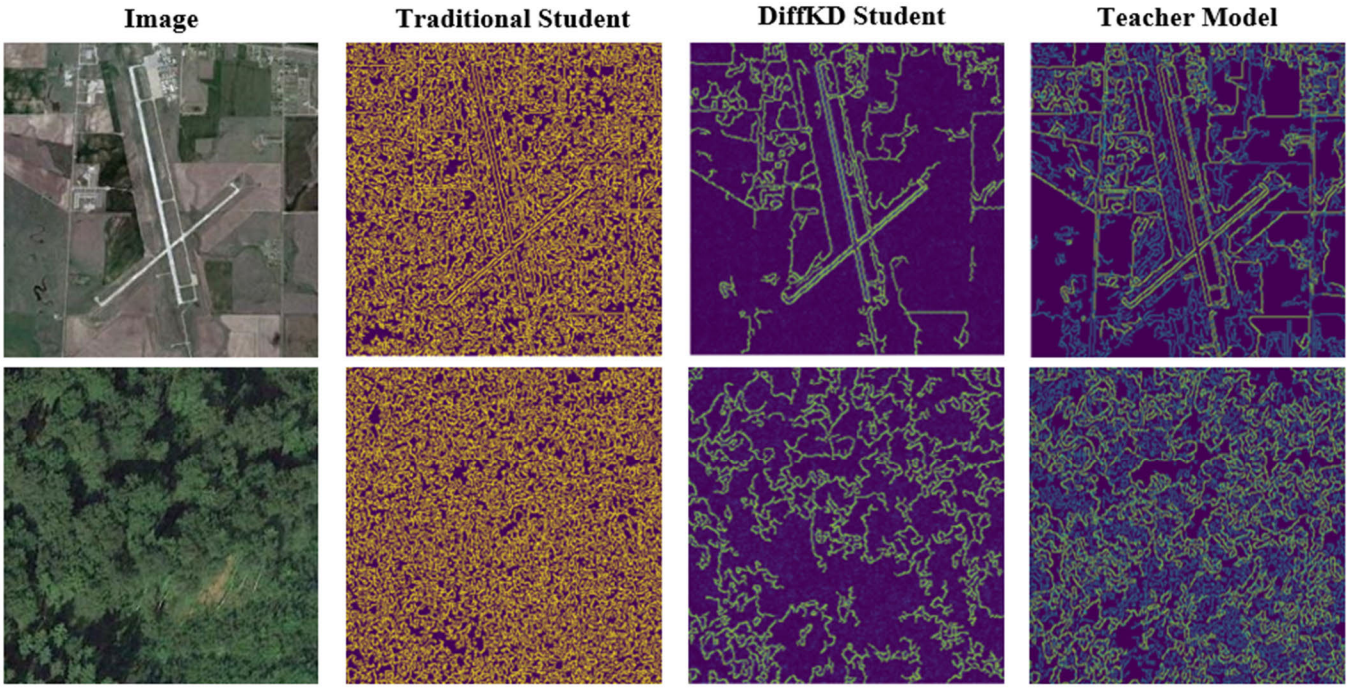


Fig. 2. Comparison of the traditional student and the DiffKD-based student in the KD framework.

and regularization (e.g., label smoothing, gradient clipping). This adapts high-level features to RS visual patterns. Post-optimization, the entire teacher model is frozen before the distillation phase commences. This ensures a stable and consistent supervisory signal throughout student training, mitigating potential instability caused by co-adaptation or teacher parameter updates.

C. Hybrid Noise Modeling for RS Physical Processes

While the DiffKD framework provides the mechanism for robust learning, its potential is fully realized only when trained on data that reflects real-world complexities. To simulate the complex degradation environment that real RS imagery faces from acquisition to processing, and to systematically evaluate and enhance the robustness of our method, we propose a comprehensive hybrid noise model that synthesizes realistic RS image degradations. Unlike naive additive noise, our model emulates four common noise sources in the RS data chain based on physical processes: sensor noise, transmission error noise, compression noise, and atmospheric disturbance noise. We introduce a global noise-level parameter $\alpha \in [0, 1]$ to control the overall noise intensity. A probabilistic mixing injection strategy is employed to realistically replicate the randomness and coupling of noise in real-world scenarios.

Sensor noise modeling aims to simulate the inherent noise introduced by the imaging device itself. We comprehensively consider three main components: photon shot noise, dark current noise, and read noise. These noise types are modeled as additive Gaussian noise, with their intensity (standard deviation) scaled linearly with the global noise level α. For a normalized input image I, the image containing comprehensive sensor noise, I_{sensor}, can be obtained by

$$I_{sensor} = clip\left(I + N\left(0, (\alpha k_{shot})^2\right) + N\left(0, (\alpha k_{dark})^2\right) + N\left(0, (\alpha k_{read})^2\right)\right). \quad (3)$$

Here, $N(0, 1)$ denotes a standard Gaussian random variable, $k_{shot} = 0.35$, $k_{dark} = 0.25$, and $k_{read} = 0.15$ are the weight coefficients for each noise component. The $clip(\cdot)$ operation ensures pixel values remain within [0, 1].

Transmission error noise modeling recreates pixel-level anomalies caused by bit errors during data transmission from the satellite to the ground station. This noise manifests as pixel values at random locations being replaced by extreme values. We first generate a binary error mask M, where each pixel location is marked as an error point with probability $p_{error} = \alpha \times 0.1$. Here, $U(0, 1)$ represents a uniformly distributed random number in [0, 1], simulating an extreme value error at that pixel location, (x, y, c) denote the spatial coordinates and channel index of the image. Subsequently, the noisy image $I_{transmission}$ is generated by

$$M(x, y) = \begin{cases} 1, & if\ U(0, 1) \le p_{error} \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

$$I_{transmission}(x, y, c) = \begin{cases} \mu(0, 1), & if\ M(x, y) = 1 \\ I(x, y, c), & otherwise \end{cases}. \quad (5)$$

Compression noise modeling simulates the blocky artifacts produced by lossy compression (e.g., JPEG) to save storage and bandwidth. We adopt a process-based approach, applying a standard JPEG compression pipeline to the original image I, with a variable quality factor E and decoder D. The compression quality factor q is negatively correlated with the noise level α, and the compressed image I_{comp} defined as

$$q = 100 - 90\alpha \quad (6)$$

$$I_{\text{comp}} = D\left(E\left(I, q\right)\right). \tag{7}$$

As α varies from 0 to 1, the quality factor q linearly decreases from 100 (near lossless) to 10 (high compression), effectively simulating visual artifacts ranging from slight to severe.

Atmospheric disturbance noise modeling simulates image haze and contrast reduction caused by atmospheric scattering and absorption. We employ a widely used atmospheric scattering physical model

$$I_{atmospheric} = I \times t + A \times (1 - t) \tag{8}$$

where $A = 0.5$ is the atmospheric light value representing haze color, set to medium gray. t is the transmission map, denoting the proportion of scene radiance reaching the sensor.

The hybrid noise integration strategy is key to simulating realistic environments. In each training iteration, noise injection is triggered with a probability $P_{mix} = 0.7$. When triggered, we randomly select 2 to 3 of the four basic noise types and apply them sequentially in a random order, thereby simulating the randomness and complexity of noise coupling. To systematically evaluate model robustness, we define three noise levels: Low ($\alpha = 0.1$), Medium ($\alpha = 0.3$), and High ($\alpha = 0.7$). During the training phase, we employ stochastic hybrid noise injection to augment the training data and improve model generalization. In the validation phase, we employ deterministic noise injection (fixing the noise types and order, and setting $P_{mix} = 1$) to establish a stable, controllable noise benchmark environment to enable a controlled and reproducible evaluation of model robustness and generalization under specific noise conditions.

D. Adaptation of DiffKD for RS Data

To make the DiffKD framework more efficient for the RS scene classification task, we implemented three key adaptations. First, to address the high dimensionality and computational cost of RS features, we introduced a linear autoencoder to compress the teacher and student features, projecting them into a low-dimensional latent space (e.g., 1024 dimensions). This significantly reduces the computational overhead of the subsequent diffusion model, enabling the entire denoising distillation pipeline to run efficiently in a single-GPU environment. Second, we introduced an adaptive noise matching (ANM) module. This module dynamically adjusts the injection level of mixed noise based on the content complexity of the input image, thereby providing the student model with more diverse noise exposure and encouraging it to learn more generalizable and robust feature representations. Finally, we conducted a trade-off study on the number of denoising steps in the diffusion model. Experiments showed that while excessive steps offer limited performance gains at significantly increased computational cost, too few steps lead to insufficient denoising. Consequently, we selected 5 denoising steps as the default setting, ensuring the generation of high-quality purified features while maintaining excellent training and inference efficiency.

III. Experimental Settings and Datasets

A. Datasets

To rigorously assess the efficacy and robustness of the proposed method, experiments were conducted on four widely-used benchmark datasets for RS scene classification: NWPU-RESISC45 (NWPU45), Aerial Image Dataset (AID30), EuroSAT (EuroSAT10), and PatternNet (PatternNet38). All datasets were partitioned into training and validation sets using a stratified random split with a ratio of 7:3, preserving the class distribution in both subsets.

NWPU45 [44] contains 31 500 images covering 45 scene categories, with spatial resolutions ranging from 0.2 to 30 m. AID30 [45] consists of 10 000 aerial images across 30 scene categories. It offers rich, detailed information with a spatial resolution between 0.5 and 0.8 m. EuroSAT10 [46] is based on Sentinel-2 satellite imagery. It comprises 27 000 multispectral images of size 64 × 64 pixels, spanning 10 LULC scene classes. PatternNet38 [47] includes 30 400 high-resolution images covering 38 scene categories. The spatial resolution varies from 0.062 to 4.693 m, containing a large number of fine-grained object classifications. Detailed information about the datasets, including resolution, acquisition time, and main scene types, is provided in Supplementary Table S1.

B. Model Configuration

Our experimental setup is based on the classic teacher-student KD paradigm. The ResNet-34 [17] architecture is selected as the teacher model due to its good balance between feature representation capacity and computational efficiency. The student model is the lighter ResNet-18 [17], which significantly reduces computational complexity by employing fewer residual blocks, making it suitable for deployment in resource-constrained scenarios.

Considering the domain gap between natural images and RS imagery, we perform targeted adaptation of the pre-trained models. First, the teacher model is adapted to the target RS dataset via a progressive fine-tuning strategy with layer freezing: initially, the backbone network is frozen, and only the classifier is trained; after the performance stabilizes, the final three layers are frozen for joint fine-tuning with the classifier. Optimization is performed using the AdamW optimizer, coupled with a cosine annealing learning rate scheduler and label smoothing to ensure training stability and generalization capability.

After this optimization, the adapted teacher model achieves an accuracy exceeding 80% on all four datasets, serving as a strong baseline and knowledge source. During the KD stage, the final three layers of the teacher model are frozen to preserve its adapted feature extraction capability while ensuring a consistent supervisory signal. The student model learns from the frozen teacher via the DiffKD framework, with hybrid noise injected during its training to enhance robustness. Specific parameter configurations are detailed in Supplementary Table S2.

TABLE I
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS

Method	NWPU45(%)	AID30(%)	EuroSAT10(%)	PatternNet38(%)
AlexNet	79.85±0.13[44]	86.88±0.56[50]	91.06[49]	93.69[49]
VGG-16	79.79±0.15[50]	86.59±0.29[48]	89.81[48]	97.5±0.02[52]
GoogLeNet	78.48±0.26[50]	86.39±0.55[45]	90.97[46]	96.12[53]
ResNet-101	87.03[51]	85.24[51]	94.74[51]	98.6±0.02[52]
SE-HRNet	93.12±0.16[54]	93.45±0.18[54]	/	/
MSTSFN	95.12±0.10[55]	**97.50±0.13**[55]	/	/
D-CNN	89.85%[56]	**97.18%**[56]	/	92.35%[56]
DMCCA + SVM	92.75%[57]	93.92%[57]	/	**99.35%**[57]
RCA	/	/	96%[58]	99%[58]
E3C	94.7%[38]	/	/	97.9%[38]
DiffKD (Ours)	**95.15%**	**94.48%**	**96.90%**	**99.12%**

C. Hardware and Software Environment

All experiments were conducted in a single-GPU environment. The PyTorch 2.7.1 deep learning framework was used with Python 3.13.5. The hardware configuration included an Intel Core i9-10900K CPU, 64 GB of RAM, and an NVIDIA GeForce RTX 2060 GPU with 10 GB of VRAM.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Performance Comparison Analysis

To comprehensively evaluate the effectiveness of our proposed method, we conducted comparative experiments between the adapted DiffKD framework and various mainstream models on four benchmark RS scene classification datasets. As summarized in Table I, DiffKD achieved the highest or highly competitive classification accuracy across all datasets, demonstrating superior generalization capability and stability.

DiffKD exhibits particularly strong performance on the large-scale and diverse NWPU45 dataset, attaining an accuracy of 95.15%, outperforming recent methods such as MSTSFN (95.12%) and SE-HRNet (93.12%). This result validates the core premise of our approach: integrating explicit feature denoising within the KD loop effectively enhances the student model's feature discriminability for complex scenes. On the AID30 dataset, DiffKD secures a robust accuracy of 94.48%, ranking among the top 3 performing methods, highlighting its consistency across varying scene distributions. For the multispectral EuroSAT10 dataset, DiffKD achieves an excellent accuracy of 96.90%, matching the performance of RCA (96%) and confirming its efficacy in leveraging joint spectral-spatial features. On the high-resolution PatternNet38 dataset, which contains abundant fine-grained details, DiffKD reaches a top-tier accuracy of 99.12%, closely approaching DMCCA + SVM (99.35%) and significantly outperforming other models such as D-CNN (92.35%). This advantage can be attributed to the diffusion process, which effectively filters out irrelevant perturbations while preserving high-frequency structural details.

Cross-dataset performance analysis highlights a significant scene dependency and a lack of generalization in traditional models. For instance, VGG-16 performed well on PatternNet38 (97.5%) but dropped significantly on NWPU45 (79.79%). Similarly, ResNet-101 excelled on EuroSAT10 and PatternNet38 but showed relatively weaker results on AID30 (85.24%). In stark contrast, DiffKD maintains leading or top-tier performance consistently across all four datasets without exhibiting any obvious weakness. This consistent high performance underscores the superior generalization capability and inherent robustness of the DiffKD framework.

To further verify the effectiveness and stability of DiffKD during optimization, we present the training curves for all four datasets in Fig. 3. The accuracy curves clearly show that the student model trained with DiffKD (blue) consistently outperforms the baseline student trained without distillation (yellow) across all datasets, achieving higher final accuracy—a direct proof of our method's efficacy. Furthermore, the DiffKD accuracy curves typically converge faster and more smoothly, especially on complex datasets such as NWPU45 and AID30. This suggests that the diffusion-based denoising process helps stabilize the training dynamics. Correspondingly, the test loss curves for DiffKD descend more rapidly, settle at lower values, and exhibit less oscillation, providing further evidence of its training stability and superior generalization.

B. Noise Robustness Analysis

To assess practical utility in real-world conditions, we systematically evaluated the robustness of DiffKD under different intensities of mixed noise (Table II). The results show that the student model exhibits varying degrees of performance degradation depending on the dataset and noise level. Under the highest noise level ($\alpha = 0.7$), the average performance drop across the four datasets is 4.13%.

The degree of degradation correlates with the intrinsic complexity of each dataset. NWPU45 and AID30, which contain semantically diverse and visually ambiguous scenes, show substantial drops of 7.37% and 8.73%, respectively. In

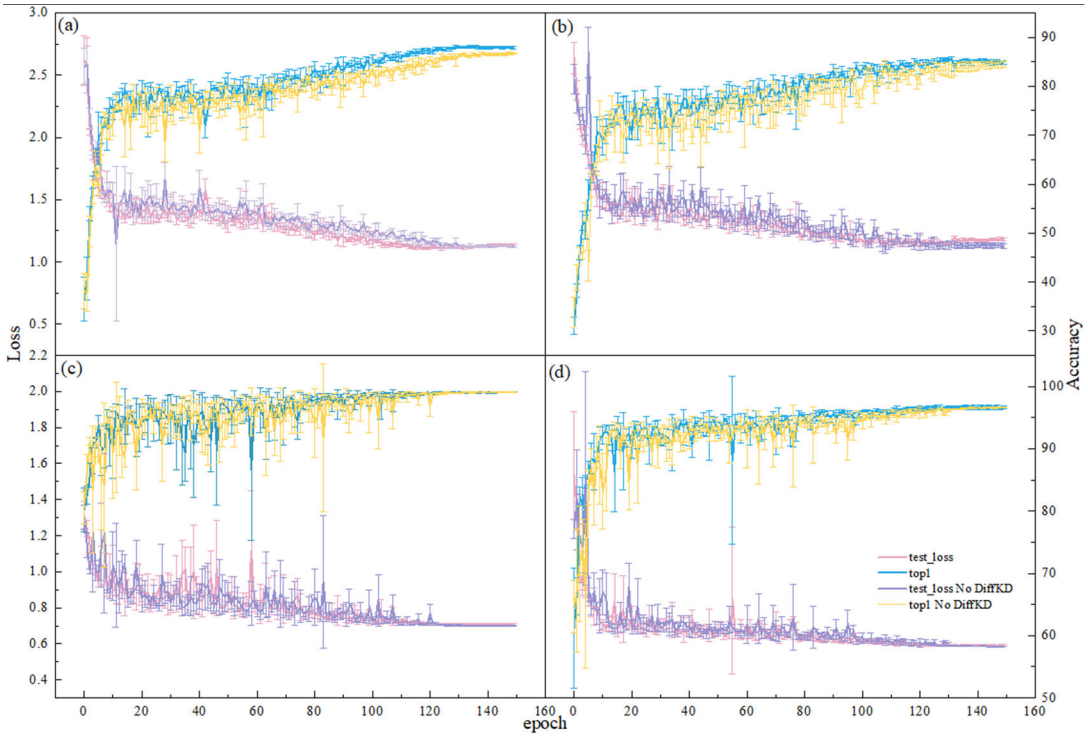


Fig. 3. Comparison of KD training processes on different datasets. (a) NWPU45, (b) AID30, (c) EuroSAT10, and (d) PatternNet38.

TABLE II
NOISE ROBUSTNESS ANALYSIS

Condition	NWPU45	AID30	EuroSAT10	PatternNet38	Noise level
Student (No Noise)	95.15%	94.48%	96.90%	99.12%	None
Teacher (No Noise)	81.48%	88.90%	92.63%	98.22%	None
Student (Noise level 1)	88.93%	86.48%	97.02%	99.18%	Low (α=0.1)
Teacher (Noise level 1)	83.93%	88.41%	92.07%	98.12%	Low (α=0.1)
Student (Noise level 3)	88.32%	86.83%	96.79%	99.25%	Medium (α=0.3)
Teacher (Noise level 3)	83.07%	87.57%	92.13%	97.93%	Medium (α=0.3)
Student (Noise level 7)	87.78%	85.75%	96.54%	99.04%	High (α=0.7)
Teacher (Noise level 7)	81.61%	85.96%	91.20%	97.32%	High (α=0.7)
Student Model Performance Drop	-7.37%	-8.73%	-0.36%	-0.08%	Average: -4.13%

contrast, EuroSAT10 and PatternNet38 demonstrate remarkable robustness, with performance reductions of only 0.36% and 0.08% under the same noise condition. This disparity can be attributed to dataset characteristics: EuroSAT10 consists of only 10 well-separated land-use classes with consistent 10 m resolution, while PatternNet38 features high-resolution images with clear object structures, enabling the model to maintain discriminative features even under strong noise. For NWPU45 and AID30, however, the combination of high intraclass variability and interclass similarity makes them more vulnerable to noise interference.

A crucial observation is that under most noisy conditions, the distilled student model not only matches but often surpasses the performance of its teacher. For example, under high noise ($\alpha = 0.7$), the student achieves accuracies of 87.78% on NWPU45 and 99.04% on PatternNet38, substantially outperforming the corresponding teacher accuracies of 81.61% and 97.32%. This serves as compelling evidence that the denoising mechanism within our distillation framework enables the student to learn noise-invariant, highly discriminative feature representations, effectively distilling a "purified" essence of the teacher's knowledge.

C. Ablation Study

To quantify the contribution of each core component in our framework, we conducted a systematic ablation study (results in Table III). Using the full DiffKD model as the baseline, we sequentially ablated four key designs: ANM, the number of denoising steps, autoencoder dimensionality, and the distillation location. A baseline without any KD was also included to measure the overall improvement.

The results distinctly highlight the importance of each component.

TABLE III
ABLATION STUDY RESULTS

Configuration	Variant	NWPU45	AID30	EuroSAT10	PatternNet38	Performance Drop
Full DiffKD		**88.93%**	**86.48%**	**97.02%**	**99.18%**	-
Adaptive Noise Matching	w/o ANM	88.92%	86.45%	97.03%	99.23%	-0.01
Denoising Steps	1-step	88.52%	86.45%	97.01%	99.21%	0.11
	10-step	88.78%	87.04%	97.07%	99.11%	-0.10
Autoencoder Dimensions	No compression	88.73%	86.24%	96.89%	99.21%	0.13
	512-dim	88.47%	86.59%	96.77%	99.12%	0.16
	2048-dim	88.43%	86.20%	96.86%	99.14%	0.25
Distillation Location	Feature only	88.55%	86.06%	96.93%	99.19%	0.22
	Logits only	88.73%	87.01%	96.79%	99.21%	-0.13
Baseline	No distillation	86.78%	84.98%	96.54%	99.04%	1.07

1) *ANM:* Removing ANM led to negligible changes across most datasets, with a slight performance fluctuation (e.g., −0.01% on NWPU45 and +0.05% on PatternNet38). This indicates that ANM, while beneficial in complex scenarios, has a mild impact under our experimental conditions.

2) *Denoising Steps Require a Balance:* A single denoising step resulted in an average performance drop of 0.11%, as insufficient purification limited the student's ability to learn robust features. Increasing the steps to ten slightly improved the average performance (+0.10%) but introduced a higher computational cost, confirming that five steps offer an optimal trade-off between efficiency and accuracy.

3) *Autoencoder Compression is Effective:* Employing no compression led to a minor average drop of 0.13%, while compressing to 512 or 2048 dimensions resulted in drops of 0.16% and 0.25%, respectively. This indicates that a moderate compression level (1024 dimensions) successfully balances information retention with redundancy removal.

4) *Feature and Logits Distillation are Complementary:* Using only feature distillation leads to an average performance decrease of 0.22%, while using only logit distillation brings an improvement of 0.13%. Although the effects of the two differ, the full model combining both achieves the highest accuracy across all datasets, demonstrating that joint distillation enables more comprehensive knowledge transfer.

All ablated variants still significantly outperformed the "no distillation" baseline by an average margin of 1.07%, solidly proving the overall effectiveness of the DiffKD framework. Importantly, the performance gain of the full model exceeds the simple sum of improvements from individual components. This suggests that the components function synergistically as an integrated whole, where each plays an indispensable role.

Additionally, we conducted ablation studies on individual noise types under a high noise level ($\alpha = 0.7$). Table S3 showed that among all individual noise components, sensor noise is the only factor that consistently causes performance degradation, while transmission, compression, and atmospheric noise each lead to performance improvement. The performance drop caused by mixed noise is far greater than that of any individual component, which validates the necessity of the multimodal noise purification mechanism within the DiffKD framework.

V. DISCUSSION AND FUTURE WORK

A. Performance Degradation Under Mixed Noise

As shown in Table II, the DiffKD student model exhibits pronounced performance degradation on AID30 and NWPU45 under mixed noise, while maintaining robustness on EuroSAT10 and PatternNet38. This discrepancy stems from dataset characteristics: NWPU45 and AID30 contain semantically ambiguous categories (e.g., church/palace, central/commercial districts) with high interclass similarity and large intraclass variance. Their wide resolution ranges (0.2–30 m for NWPU45, 0.5–0.8 m for AID30) and prolonged acquisition periods further introduce uneven image quality and illumination variations. Mixed noise amplifies these issues, blurring fine-grained discriminative features and increasing intra-class variability, thereby impairing classification. In contrast, EuroSAT10 (uniform 10 m resolution, stable imaging conditions) and PatternNet38 (high-resolution, distinct object structures) exhibit stronger noise resilience due to their consistent quality and clear class boundaries.

B. Effects of Individual Noise Types

When applied individually, only sensor noise consistently degrades performance (e.g.,–0.56% on AID30), as it directly corrupts fine details. In contrast, transmission, compression, and atmospheric noise yield slight performance gains across datasets (e.g., +2.29% on AID30 for transmission noise), acting as an implicit data augmentation that mitigates overfitting. This contrast underscores the unique detrimental effect of sensor noise versus the beneficial regularization provided by other noise types.

C. Limitations and Future Work

Current noise modeling focuses on four typical RS noise types; future work should incorporate more complex real-world degradations such as shadow interference. The number of denoising steps remains empirically set—adaptive strategies could further improve efficiency. We also plan to extend the denoising distillation framework to other tasks like object detection and semantic segmentation to assess its broader applicability.

VI. Conclusion

In this work, we demonstrated that integrating an explicit feature denoising process via diffusion models into the KD framework effectively addresses the dual challenges of domain shift and mixed-noise degradation in RS scene classification. Our adapted framework, supported by a tailored teacher optimization strategy and a realistic hybrid noise model, achieves state-of-the-art accuracy (e.g., 96.90% on EuroSAT10) and exceptional robustness. Notably, the compact student model surpasses its teacher in noisy conditions, proving its ability to learn superior, noise-invariant representations suitable for edge deployment. Collectively, these results underscore the potential of diffusion-based denoising distillation for robust and efficient RS scene classification in real-world applications.

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